

Dear Editor,

We submit our manuscript, “Adversarial Vulnerability of Machine Learning GNSS Spoofing Detectors to Transmitted Spoofing Signals,” for consideration as an Original Article in *Satellite Navigation*.

Relationship to a previous submission

An earlier manuscript by the same authors on learned spoofing detectors (SANA-D-26-00339) was declined by *Satellite Navigation* in September 2026. We submit the present work as a new manuscript rather than a revision, because the study it reports is different in method and in evidence.

The earlier submission evaluated detectors by perturbing receiver observables directly and argued separately that the resulting perturbations were physically attainable. The reviewer identified that argument as the weak point of the study. In the present work the attack is constructed as a radio waveform: a counterfeit GPS L1 C/A signal is synthesised at sample level, summed with authentic recordings, and tracked by an open software receiver, so the observables each detector reads are produced by receiver processing of a transmittable signal. The question of realizability is therefore settled by the construction of the experiment rather than by an argument attached to it. The results, the figures and the conclusions are new.

Why this manuscript belongs in *Satellite Navigation*

Machine learning spoofing detectors are moving from research prototypes towards integration in receivers, and their reported accuracy is obtained against fixed corpora of recorded attacks that do not respond to the detector under test. This manuscript evaluates fourteen detectors, eight classical and six deep, against 126 transmitter settings that satisfy the published conditions for receiver takeover, with every detector held at a common operating point and the false alarm rate reported alongside every result.

Three results follow. Clean detection performance predicts resistance poorly, and at the top of the ranking the two orderings are close to inverted. Where resistance appears it is purchased with false alarm rates of 46 to 74 percent on authentic data, which is not operationally available. A missed attack propagates to the navigation solution, driving the receiver clock at 48.89 nanoseconds per second against a commanded 48.87.

The work addresses resilient positioning, navigation and timing, a core topic of the journal, and connects directly to the literature on learning based GNSS spoofing detection the journal has published. The library of transmitted attacks is a reusable artefact for evaluating physics constrained and certified detectors.

Policy issues

We are not aware of any policy issues relevant to this submission.

Competing interests

The authors declare that they have no competing interests.

Prior publication and authorship

Beyond the declined submission described above, this manuscript has not been published and is not under consideration for publication elsewhere, in whole or in part. All authors have seen and approved the manuscript in its current form and agree to its submission to *Satellite Navigation*.

Special issue

This submission is not intended for a special issue.

Thank you for considering our manuscript.

Sincerely,

Wasiu Akande Ahmed and Joel Segun Ojerinde (corresponding authors)

On behalf of all authors

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